COMS 4172: 3D User Interfaces and Augmented Reality

Mobile GamAR Written Documentation

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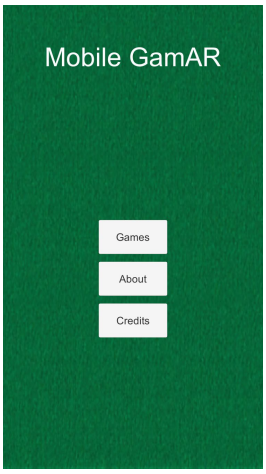
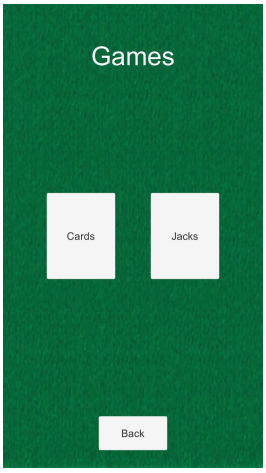
Main Menu

The main menu of Mobile GamAR allows users to obtain information about the application and start the supported games.



Games

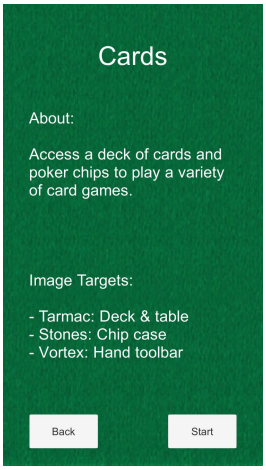
The games window allows users to pick which supported game to play.



Cards

The Cards menu gives users information on the virtual cards scene as well as a list of Image Targets needed to use the scene. The following are the needed Image Targets all of which were provided by instructors:

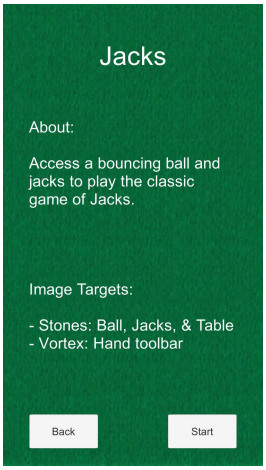
1. Tarmac: Deck & table
2. Stones: Chip case
3. Vortex: Hand toolbar



Jacks

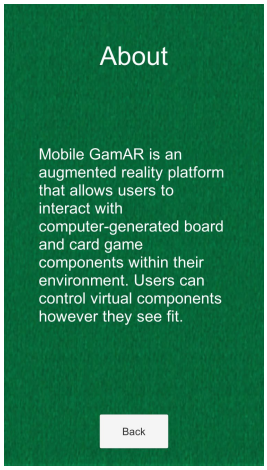
Identical to the Cards menu, Jacks menu gives users information on the virtual cards scene as well as a list of Image Targets needed to use the scene. The following are the needed Image Targets all of which were provided by instructors:

1. Stones: Ball, jacks, & table
2. Vortex: Hand toolbar



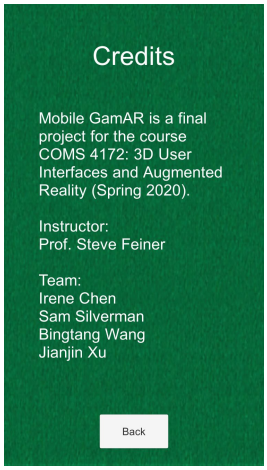
About

The about menu gives general information on Mobile GamAR and the goals of the application.



Credits

The credits menu lists both the course, instructor, and team members that led to the creation of Mobile GamnAR.



Game: Playing Cards

Game Object: Cards

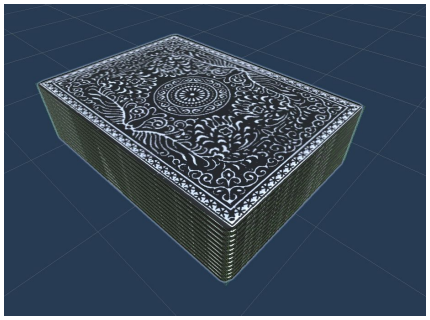
The game consists of the standard 52 cards in a playing card deck.

Each card supports 4 modes of manipulation:

1. Move: A card can be moved to any position in the game.
2. Rotate: A card can be rotated to any desired orientation.
3. Discard: A card can be discarded back into the deck.
4. Flip: A card can be flipped over.

Game Object: Deck

The deck is initialized with a set of 52 cards stacked and laid face down on the center of the game board. The deck originates each of the dealt cards.

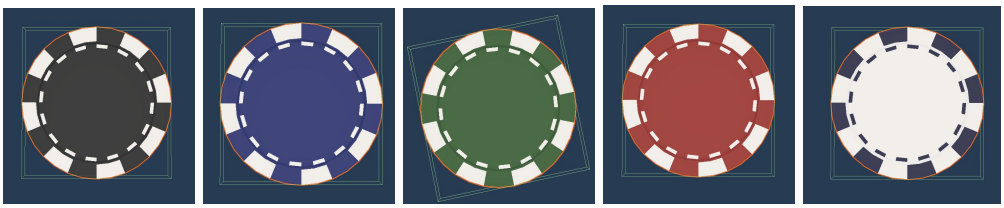


The deck supports 5 modes of manipulation:

1. Move: The deck can be moved to any position in the game.
2. Rotate: The deck can be rotated to any desired orientation.
3. Shuffle: The deck can be shuffled.
4. Deal: The deck can deal the top card.
5. Collect: The deck can collect all dealt cards in the game.

Game Object: Chips

The game supports usage of 5 chip types: black, blue, green, red, and white.

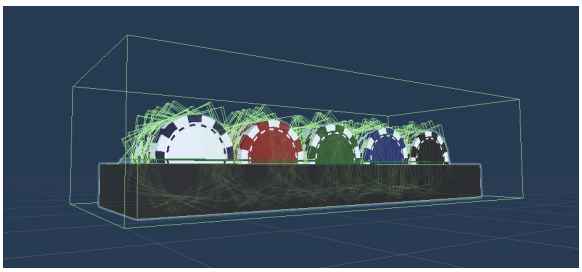


Each chip supports 2 modes of manipulation:

1. Move: A chip can be moved to any position in the game.
2. Discard: A chip can be discarded back into the chip case..

Game Object: Chip Case

The chip case is initialized upon detecting the stones image target. The chip case is used to manage all chips within the game. The chip case spawns new chips that can be used by a player. The user can obtain chips from the chip case (as well as discard chips back into the chip case) by indicating a quantity of each chip needed using + - incrementers and then choosing to either deal the chip amount or discard the chip amount.

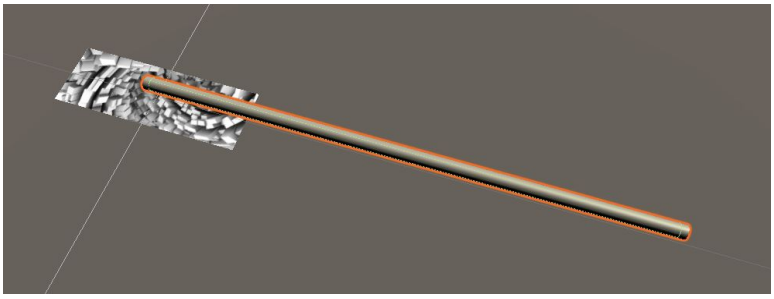


The chip case supports 2 modes of manipulation:

1. Deal: The chip case can deal new chips to the game.
2. Collect: The chip case can collect all dealt chips in the game.

Toolbar

A hand toolbar (represented as a wand) is used for game selection and manipulation. The toolbar is initialized upon detection of the vortex image target.



To begin interacting with a game object, the user must select said object. This is achieved by touching the toolbar hand to the game object.

UI Triggers

When the toolbar selects a game object, the game object's manipulation UI is displayed. This UI contains buttons that navigate users between the available manipulation modes the game object supports. Every manipulation UI contains a done button which, when pressed, deselects the game object and disables the manipulation UI.

Wayfinding

Game Bots

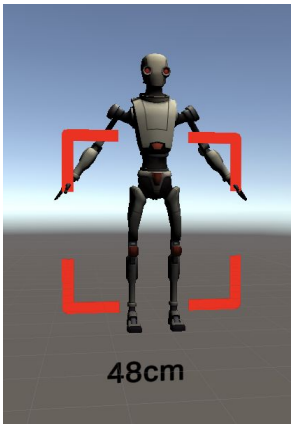
The game supports bot players. The bot players walk randomly (wander) around an edge of the game table and represent locations and movements users would expect when playing in-person with 1+ other individual.

Wayfinding: Minimap

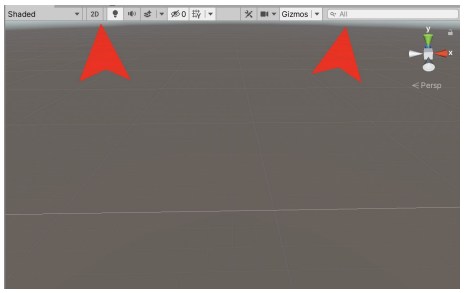
The game includes a minimap at the top right corner of the screen. The minimap enables the user to visually localize themselves amongst other (bot) players by providing a top down allocentric view of the game. The player's position within the game world is identified by a blue dot icon. Adversarial opponents (bots) are identified with a red dot icon on the minimap. The minimap follows the player's game coordinates as they navigate around the game.

Wayfinding: Targets and Offscreen Indicators

Each adversarial opponent visible within screen bounds is identified with a red box target. The box target is accompanied by a distance measurement (in cm) that indicates the opponent's distance relative to the player.



Players that exist but are not within the view of the screen bounds (off screen opponent) are located with an arrow pointing in the direction of the figure. These arrows are also accompanied by a distance measurement (in cm) that indicates the opponent's distance relative to the player.



Playing Cards Design Rationale

We wanted to support a variety of card games that all use the same pieces (deck and chips). To do this, we only support the basic functionalities one would perform on these game pieces (move, rotate, flip, shuffle, etc.). This gives users the freedom to play whatever and however they want.

The cards are a part of the game's main image target since all card games obviously require cards. We felt it was important to give the chip case and chips their own image target so users can choose to add them to their game or not by simply showing or hiding the image target.

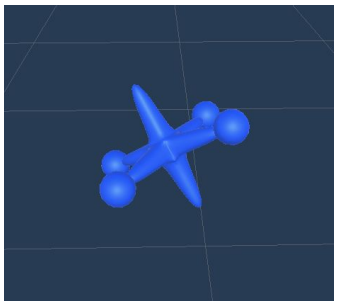
Additionally, we felt that because card games are inherently played with your hands, a hand toolbar provided the most intuitive and easy approach to selecting and manipulating game pieces. For simplicity we made the hand toolbar a want and for ease of use, we made the want long to simulate a user's reach within the game.

Finally, because many card games require awareness of where your opponents are sitting, we provided full functionality for wayfinding to easily see opponents. This aids in games like Poker which rely on knowledge of opponents and what they are doing.

Game: Jacks

Game Object: Jacks

Ten jacks are initialized in a pile at the center of the game board.



The jacks have different modes of manipulation depending on if a current jacks turn is active or not. If a jacks turn is not active, then the jacks support 2 modes of manipulation:

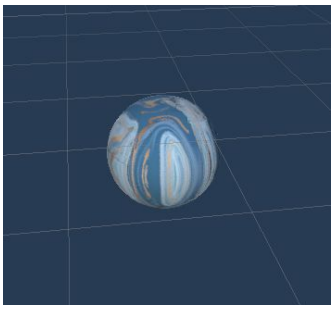
1. Collect: All jacks are collected and placed in the user's hand.
2. Drop: All jacks are dropped from the user's hands. If a jack drops off of the game board, the jack is re-initialized to the pile where it was found at the beginning of the game.

If a current jacks turn is active, the jacks support 1 mode of manipulation:

1. Pick Up: Individual jacks are collected and placed in the user's hand.

Game Object: Ball

A ball is initialized at the center of the game board.



The ball supports 2 modes of manipulation:

1. Collect: The ball is collected and placed in the user's hand.
2. Drop: The ball is dropped and bounces on the game board. If the ball drops off of the game board, the ball is re-initialized to the position it was found at the beginning of the game.

If a current jacks turn is active, the ball supports 1 mode of manipulation:

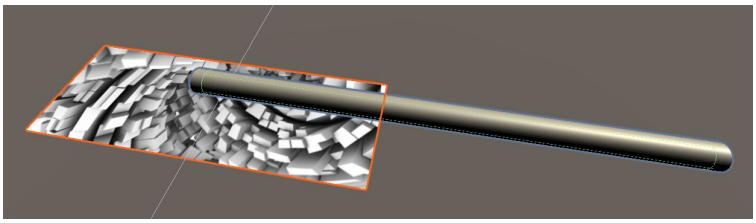
1. Pick Up: The ball is picked up and the ball along with all jacks are re-initialized in their default positions.

Starting a Jacks Turn

A jacks turn is started when a user drops the ball. During a jacks turn, jacks and the ball are automatically picked up upon selection. A counter is displayed showing how many jacks the user has picked up. The jacks turn ends when the user picks up the ball in which the jacks and ball are returned to their default positions.

Toolbar

Much like the toolbar for the Playing Cards Game, a hand toolbar (represented as a wand) is used for game selection and manipulation. The Jacks hand toolbar is slightly shorter than the Playing Cards hand toolbar. The toolbar is initialized upon detection of the vortex image target.



To begin interacting with a game object, the user must select said object. This is achieved by touching the toolbar hand to the game object.

UI Triggers

When the toolbar selects a game object, the game object's manipulation UI is displayed. This UI contains buttons that navigate users between the available manipulation modes the game object supports. The user can switch between the ball and jacks by touching the two objects and the UI adjusts. During a jacks turn, the toolbar automatically picks up game objects without needing additional button input from the user.Travel
When the stones image target is active and displays the game, two buttons appear on the bottom left and right of the screen.

These buttons allow the user to virtually move left and right around the game. These buttons allow the user to virtually travel in the game environment to obtain better angles to make jacks game moves. Although it is possible for the user to physically walk around the game, this virtual travel allows for easy mobility when the user may not always be able to conveniently move.

Jacks Design Rationale

We wanted to make playing jacks easy and convenient for users so unlike the Playing Cards scene, the number of available actions for users is limited. Users are able to pick up and throw the jacks and ball which are the two actions in the game. To make collecting the jacks to throw easier, users are able to select any jack and can collect them all at once. Throwing the jacks and ball is also made easy as the user pushes a button to drop whichever item they are holding in their hand.

Additionally, in case the user drops something off the surface of the game, a respawn zone will move whatever falls off back to its default position. This keeps all the game objects on the game board.

When the ball is bouncing and the jacks turn is in progress, we wanted to both make picking up jacks easy and obvious. Users simply have to touch a jack to pick it up and a counter displays how many have been picked up. No additional input is needed. To end the jacks turn, the user has to touch the ball to pick it up. Doing so resets all pieces to their default position allowing the user to play again.

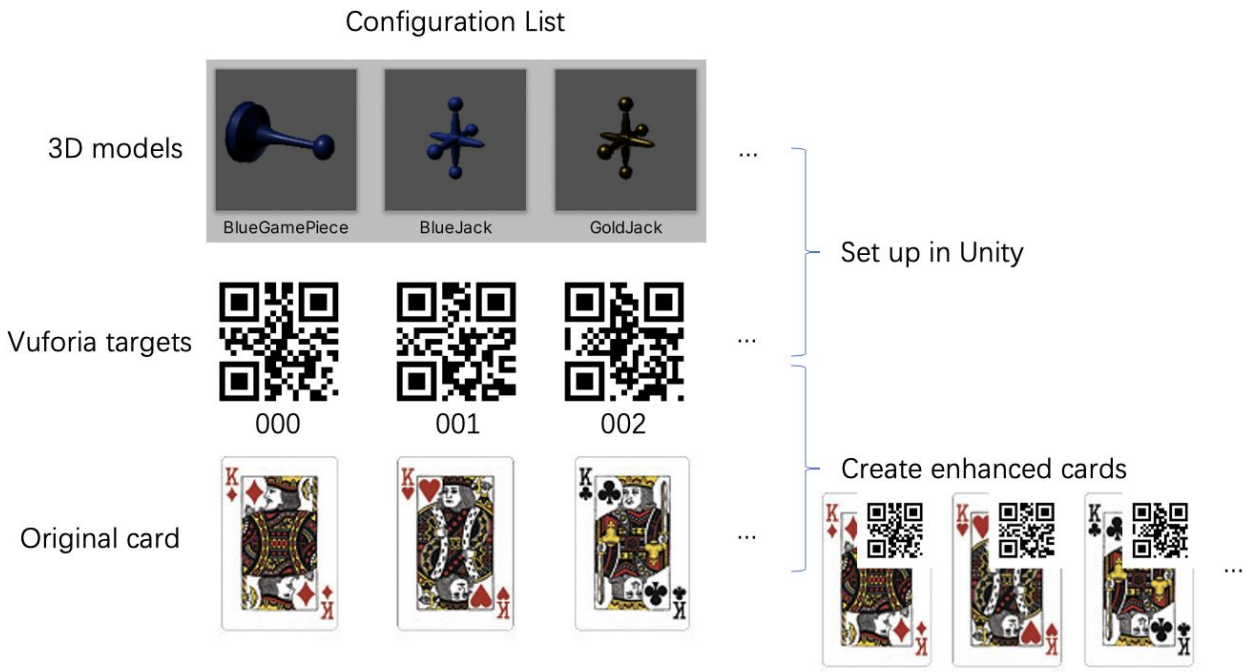
Similar to Playing Cards, we felt that because jacks is played with your hands, a hand toolbar provided the most intuitive and easy approach to manipulating game pieces. This hand toolbar is slightly shorter so users have to move to grab pieces.

Finally, because one's position around the game greatly alters their turn, we have provided a virtual method for travel within the game. Buttons move the user left and right around the game to change position. While we acknowledge that a user could simply walk around the environment, providing a virtual method for travel ensures that a user will always be able to move even when their physical environment prevents them from doing so.

Toolkit to Augment Card Games

When users are not familiar with 3D interactions, or we have not supported the game users want to play, it is also plausible to play with 2D cards but view it in a fancy 3D way. It would be interesting for players to see 3D characters on their cards.

The user only needs to give a list specifying the connection of 3D augmentation objects with Vuforia image targets. The list is composed of triplets (target name, 3d model name, card images). As shown in the pipeline below, the list specifies how a Vuforia target is matched to desired 3D models. We also provide a script for composing original card images and Vuforia targets together to form enhanced cards.



Considering this persona: A game developer designed a custom D&D card game and wants to enhance the game with 3D views because he thinks it would be great if the characters and monsters are displayed in 3D. Using our toolkit, he first finds a list of 3D models. He decides to use our default QRCode Vuforia targets so he can avoid finding his own image targets. Second, he writes a list specifying which card is matched to which target and model. Third, he imports this configuration list into the Unity project and the building succeeds. Lastly, he runs another script to obtain the enhanced cards. This is all the steps for enhancing a card game.

Our toolkit is essential because otherwise the game developer would need to create hundreds of image targets in Unity repeatedly and append the corresponding 3D models to image targets. With our toolkit, only a list is required.

It is worth noticing that QRCode is not a good Vuforia target, because the difference between different targets is too subtle. This will introduce tracking difficulty. However, it is not hard to change to other images. We provide a code to upload batches of images as Vuforia targets.

PokerConfig.txt

```
0 Prefabs/GamePeices/BlueGamePiece QRCode/imgs/000.png
1 Prefabs/GamePeices/BlueJack QRCode/imgs/001.png
2 Prefabs/GamePeices/GoldJack QRCode/imgs/002.png
3 Prefabs/GamePeices/GreenJack QRCode/imgs/003.png
4 Prefabs/GamePeices/RedGamePiece QRCode/imgs/004.png
5 Prefabs/GamePeices/RedJack QRCode/imgs/005.png
6 Prefabs/GamePeices/SilverJack QRCode/imgs/006.png
7 Prefabs/GamePeices/YellowGamePiece QRCode/imgs/007.png
8 Prefabs/GamePeices/YellowJack QRCode/imgs/008.png
9 Prefabs/GamePeices/BlueGamePiece QRCode/imgs/009.png
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